Cross-Task Recurrence of a Candidate Rule-Sensitive Neural Direction as a Predictor of Later Bedside Command-Following

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Abstract

Behavioral nonresponse after severe brain injury can reflect impaired consciousness, impaired sensory or language access, inability to express a command through movement, fluctuating arousal, or failure of a neural test. Active electroencephalographic and functional MRI paradigms can reveal covert command-related responses in some behaviorally nonresponsive patients, but existing evidence also demonstrates task and modality dissociations and frequent false negatives [1] [4] [25]. Clinical guidelines consequently support serial standardized examination and multimodal assessment rather than reliance on a single observation [3] [15].

We propose a multicenter longitudinal study of adults who, at 10–14 days after an acute acquired brain injury, meet behavioral criteria for unresponsive wakefulness syndrome or minimally conscious state minus and do not show reliable bedside command-following. The primary empirical target is **recurrent rule-sensitive neural responding**, not phenomenal consciousness. Participants will complete motor-imagery and spatial-navigation tasks in EEG and fMRI at approximately 14, 21, and 28 days after injury. Arbitrary auditory cues will receive counterbalanced, mini-block-level mappings to assigned perform and assigned withhold conditions. An active nonimagery control, passive replay, objective auditory checks, receptive-language proxies, electromyography, pupillometry, medication measurement, and lesion-aware physiological processing will constrain sensory, linguistic, motor, vigilance, and generic-preparation alternatives.

The original modality-specific latent rotations are replaced by an identifiable external reference. A fixed fMRI-anchored parcel-by-phase coordinate system will be learned in independent calibration data without patient outcomes or imagery perform-versus-withhold labels. The EEG-to-reference encoder will have a uniquely specified regularized objective; its orientation, signs, coordinate order, and hyperparameters will be frozen across participants, sessions, sites, and folds. New patients will receive anatomical registration, condition-blind centering, and positive scaling only. Patient-specific rotations, reflections, sign choices, and contrast-optimized alignment are prohibited.

The Anchored Recurrent Instruction-Sensitivity score (ARIS) will test a complete, connected graph of the four task-by-modality cells. It includes same-task cross-modal, cross-task within-modal, leave-one-task, leave-one-modality, leave-one-cell, and cross-session transport tests. Every task-by-modality cell in a qualifying session pair must support the same signed perform-minus-withhold direction; a favorable average cannot conceal an opposed cell. Low-norm contrasts are regularized and must also pass an externally frozen directional-reliability threshold. Recurrence requires one qualifying pair among the three prespecified early sessions, with pair selection included inside the null calibration.

The primary patient-level probability is an exact or Monte Carlo design-based randomization probability generated by reconstructing the restricted mini-block allocation mechanism. The complete label-adaptive pipeline is rerun for each assignment. Temporal shifts and phase-randomized surrogates are retained only as negative-control distributions, not represented as exact tests. The primary day-28 landmark predicts study-defined reliable CRS-R command-following by 180 days after injury, under observed care. Confirmation requires both the lower one-sided 97.5% confidence bound for external AUC to exceed 0.80 and the corresponding lower bound for paired incremental AUC to exceed 0.10 over a penalized clinical, structural, resting-network, acquisition-quality, and single-task baseline.

A positive result would support a prognostic association between recurrent rule-sensitive neural responding and later bedside command-following. Stronger interpretations involving covert command-following capacity, limited agency, access consciousness, or phenomenal consciousness would remain defeasible and would require convergent sensory, language, content, and clinical evidence. A negative result would not establish absence of consciousness. Failure of estimator calibration, control-cohort sensitivity, temporal prediction, external discrimination, or incremental value would defeat the corresponding protocol claim rather than being redescribed as hidden success.

Keywords: disorders of consciousness; cognitive motor dissociation; command-following; EEG; functional MRI; motor imagery; spatial navigation; neural geometry; prognosis; dynamic landmarks; thalamocortical networks

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1. Research Question, Construct Boundaries, and Study Status

1.1 Primary research question

The primary question is:

> **Among adults without reliable behavioral command-following at 28 days after acute acquired brain injury, does recurrent, connected, rule-sensitive neural responding across motor imagery, spatial navigation, EEG, fMRI, and at least two early sessions predict the later emergence of reliable bedside command-following by 180 days after injury?**

The question is intentionally narrower than whether a patient is conscious. It concerns an observable assignment-sensitive neural relation and a future behavioral endpoint.

The primary construct is named **Anchored Recurrent Instruction-Sensitivity** (ARIS). “Anchored” means that both modalities are expressed in a common coordinate system whose orientation is fixed using independent data. “Recurrent” means that the relation must be present in two prespecified early sessions and must transport between them. “Instruction-sensitivity” means that neural activity follows the currently assigned perform-versus-withhold relation rather than one physical cue. The name does not assert that a private intention, imagery experience, or conscious act has been directly observed.

1.2 Primary hypothesis

Among participants alive, observed, not already meeting the behavioral endpoint, and technically evaluable at the 28-day landmark:

1. continuous ARIS will predict reliable bedside command-following between day 28 and day 180 after injury;
2. a frozen model adding ARIS will achieve an external-site AUC whose lower one-sided 97.5% confidence bound exceeds 0.80; and
3. the paired increase in AUC over the frozen complete baseline model will have a lower one-sided 97.5% confidence bound exceeding 0.10.

Both discrimination requirements must be met. A point estimate just above either threshold is not confirmatory if its lower confidence bound remains below the threshold.

1.3 Secondary hypotheses

The study has four secondary hypotheses.

1. **Temporal hypothesis:** at fixed day-28, day-60, and day-90 risk sets, ARIS estimated exclusively from prior data will predict subsequent command-following.
2. **Content hypothesis:** some ARIS-positive participants will also express independently frozen motor-imagery and navigation-content templates, strengthening—but not proving—the inference that assigned imagery was implemented.
3. **Network-support hypothesis:** longitudinal changes in selected alpha-network and thalamocortical measures will precede changes in ARIS more strongly than the reverse temporal ordering.
4. **Dissociation hypothesis:** ARIS will add prognostic information beyond isolated task decoding, while isolated task responses will remain more prevalent and less specific.

1.4 Distinct variables that must not be collapsed

The protocol separates the following variables.

Variable	Operational meaning in this study
Clinical state classification	UWS or MCS− classification based primarily on serial standardized behavioral assessment and documented multimodal clinical information.
Auditory access	Evidence that task sounds reach the nervous system, assessed through clinical examination and objective auditory-pathway measures where feasible.

Variable	Operational meaning in this study
Receptive-language access	Evidence that spoken instructions are processed beyond elementary acoustics, assessed through history, examination, and an independent neural language-control paradigm.
Instruction-assignment effect	A neural cue-by-mapping interaction identified by the randomized schedule.
Rule-sensitive neural responding	Reproducible instruction-assignment effects that follow cue meaning rather than physical cue identity.
Imagery-content support	Alignment with independently frozen motor-imagery or navigation-specific neural templates.
Motor expression	Capacity to produce a publicly observable movement in response to a command.
Reliable bedside command-following	The study-defined repeated CRS-R endpoint specified below.
Communicative report	Reliable intentional selection among alternatives sufficient to convey a message. ARIS is not a communication test.
Access consciousness	A theoretical capacity for information to be broadly available for reasoning, control, or report. It is not directly measured by the endpoint.
Phenomenal consciousness	Whether there is something it is like for the patient. It remains underdetermined by this protocol.
Agency	Intentional authorship or selection of action. The experiment tests compliance with an externally assigned rule and therefore supports, at most, a limited and defeasible agency inference.

These distinctions prevent a future behavioral outcome from being mislabeled as "recovery of consciousness." The validated endpoint, if validation succeeds, will be **emergence of reliable bedside command-following under observed care**.

1.5 What a positive result would establish

A positive primary result would justify the statement:

> In the defined acute/subacute, MRI-complete population, a connected and recurrent neural response to assigned perform-versus-withhold rules predicts the later emergence of reliable bedside command-following beyond prespecified clinical, structural, resting-network, acquisition-quality, and isolated-task information.

It would not establish:

- a particular phenomenal experience;
- intact language comprehension in every positive participant;
- deliberate imagery or deliberate withholding without content support;
- autonomous goal selection;
- an intrinsic probability of recovery independent of treatment;
- a universally transportable clinical test;

- that ARIS is necessary or sufficient for consciousness.

1.6 What a negative result would establish

A technically valid negative result would show that the protocol did not detect the specified recurrent relation. It would not show that the patient lacks awareness, thought, pain, language, memory, or all forms of command comprehension. A protocol that depends on auditory access, rule retention, two imagery abilities, two measurement modalities, and repeated recording has many false-negative pathways.

A distinction will therefore be maintained among:

1. **ARIS positive;**
2. **ARIS negative under technically adequate measurement;**
3. **directionally weak or indeterminate;**
4. **technically invalid;**
5. **not observed because of death, withdrawal, instability, or unavailable modality.**

1.7 Pre-preregistration status

This revision resolves the formal arbitrary-rotation and disconnected-pairing defects in the earlier estimator at the level of algorithmic specification. It does not supply the empirical simulation and control-cohort evidence needed to show that the replacement is adequately sensitive, calibrated, or physiologically meaningful. The study is therefore not ready for confirmatory registration or patient recruitment.

Before recruitment, Stage 0 must demonstrate:

- recovery of a planted common direction;
- rejection of checkerboard alternatives;
- resistance to arbitrary mapping perturbations;
- calibrated design-based false-positive rates under autocorrelation and missingness;
- adequate healthy and responsive-control sensitivity;
- low passive-replay positivity;
- acceptable transport of the external EEG-to-reference encoder;
- sufficient power for both external AUC criteria under site heterogeneity.

Failure of any mandatory Stage 0 gate requires redesign or abandonment before patients are exposed to research procedures.

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2. Clinical and Scientific Background

2.1 Behavioral nonresponse has multiple causes

Severe brain injury can impair arousal, awareness, sensory access, language comprehension, memory, motor planning, corticospinal output, or combinations of these systems. Bedside nonresponse consequently does not identify one neural condition. Reviews of disorders of consciousness describe preserved neural networks and covert task responses in some patients who appear behaviorally unresponsive and estimate cognitive motor dissociation in up to approximately 15–20% of affected patients [1]. That estimate supports the clinical importance of searching for dissociations; it does not prove that every active-paradigm response is conscious command-following.

Serial standardized examination remains foundational. European guidance recommends comprehensive neurological examination, repeated CRS-R assessment in subacute and chronic settings, standard EEG, evaluation of sleep patterns, and consideration of active or resting EEG and functional neuroimaging where feasible [3]. The same guidance recommends the FOUR score rather than the Glasgow Coma Scale for acute assessment [3]. American practice guidance likewise recommends treating confounding conditions, optimizing arousal, and performing serial standardized assessments [15].

This protocol therefore uses serial CRS-R and FOUR assessment rather than treating neural testing as a replacement for clinical examination.

2.2 Command-following is temporally unstable

One behavioral observation can be misleading. A 2026 preprint record using the Glasgow Coma Scale motor item reported that first documented command-following was transient in 22.2% of recovering patients and that repeated threshold crossing was common; sedation was strongly associated with non-following [18]. That record is not direct validation of the proposed CRS-R endpoint, and its publication status must be verified before final registration. Its endpoint also differs from the full CRS-R. It nevertheless motivates repeated assessment and explicit sedation measurement.

The present study does not describe a first command response as a stable state. The endpoint requires confirmation by another examiner on another day.

2.3 Active neural paradigms are informative but imperfect

Appropriate imagery-related activity following an instruction is harder to explain by passive stimulus processing than a simple sensory response. Even so, a single positive analysis can arise from chance, artifacts, cue identity, mapping reminders, differential expectancy, generic preparation, or analytic adaptation. A negative analysis can arise from hearing loss, receptive-language dysfunction, fatigue, task-specific injury, poor imagery ability, movement, altered neurovascular coupling, or insufficient signal.

Task and modality dissociations have been observed. In one study, different behaviorally nonresponsive patients showed evidence in spatial-navigation fMRI, motor-imagery EEG, or both motor-imagery modalities, while complete multimodal data were available for only six participants [4]. This supports the use of multiple tasks and measurements. It does not demonstrate that a single shared task-general direction exists.

False negatives also occur among people who behaviorally follow commands. In a small fMRI comparison, motor-imagery detection varied by paradigm, and some behaviorally command-following patients were not detected by the imagery test [25]. A negative ARIS result must consequently remain nonexclusionary.

2.4 Why recurrence and connected generalization may help

Multiple tests can be combined in two fundamentally different ways.

A **union rule** declares a participant positive if any task or modality is positive. This can improve sensitivity but also permits one task-specific or artifactual result to determine classification.

A **connected recurrence rule** requires that the same signed relation survive:

- motor imagery and spatial navigation;
- EEG and fMRI;
- same-task cross-modal tests;
- cross-task within-modal tests;
- leave-one-task and leave-one-modality tests;
- more than one session.

The proposed rule is more demanding and may be less sensitive. Its intended advantage is that no isolated task-by-modality cell can establish positivity.

2.5 Resting networks as comparators, not assumed mechanisms

Resting EEG studies support the relevance of large-scale network organization. In a study of 32 patients with chronic disorders of consciousness, alpha-band networks showed reduced local and global efficiency, fewer hubs, and spatially restricted modules; alpha-network efficiency correlated with behavioral awareness, and some behaviorally unresponsive patients with active-neuroimaging evidence had relatively preserved alpha networks [2].

Network metrics are not uniformly monotonic indicators of favorable function. Another study reported reduced interhemispheric frontoparietal connectivity in UWS relative to MCS but also higher clustering and local efficiency for some measures in UWS [34]. The protocol will therefore not combine every "higher" graph value into a generic better-network score. Metric direction and biological interpretation will be specified separately.

A preprint combining resting fMRI and clinical features reported approximately 90% outcome discrimination across three datasets from two centers [33]. That result is preprint-level evidence, and accuracy is not interchangeable with AUC, calibration, or locked prospective validation. It nevertheless requires the present study to compare ARIS against a serious multidomain baseline rather than a minimal clinical model.

2.6 Thalamocortical and anterior forebrain proposals

A neuroimaging review emphasizes that isolated regional activation is insufficient and highlights thalamocortical circuits plus differentiated connectivity within and between networks [35]. Model-dependent resting-fMRI work has reported altered coupling within the anterior forebrain mesocircuit and between that system and the default-mode network in prolonged disorders of consciousness [38]. Because those directed parameters depend on a generative model, they are not direct observations of causal traffic.

The present mechanistic proposal is modest: selected thalamocortical and long-range cortical properties may make stable rule-sensitive responding more likely. They are not treated as established necessary or sufficient conditions.

2.7 Treatment and observed-care prognosis

Clinical recovery depends on treatment, complications, rehabilitation, sedation, stimulation, and care-limitation decisions. Practice guidance supports amantadine for a defined subgroup of adults with traumatic disorders of consciousness 4–16 weeks after injury [15]. Protocols have proposed apomorphine and transcutaneous vagal stimulation as interventions affecting anterior forebrain or thalamocortical systems, but the cited records are trial protocols rather than efficacy evidence [36] [37].

The primary estimand is therefore prognosis **under observed care**. It is not intrinsic biological recoverability and not the outcome under a hypothetical uniform treatment policy.

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3. Operational Framework

3.1 “Endogenous” is not used as an endpoint label

The earlier manuscript used “endogenous control” too strongly. Every trial begins with an external cue, and the experiment does not directly observe a private intention. The revised protocol uses “rule-sensitive neural responding.”

Consider two tones, $u = -1$ and $u = +1$. In one mini-block, the high tone means assigned perform and the low tone assigned withhold. In another, the mapping reverses. A neural effect that follows the high tone is tied to physical cue identity. A neural effect that reverses with the assigned mapping is sensitive to the cue-by-context rule.

That interaction can still reflect:

- conditional auditory association;
- retrieval of the current mapping;
- expectancy;
- response preparation;
- effort;
- inhibition;
- nonconscious rule-conditioned processing.

Cue reversal excludes a fixed-cue account but does not uniquely identify deliberate imagery, deliberate withholding, agency, or awareness.

3.2 Assigned perform and assigned withhold

The terms **assigned perform** and **assigned withhold** describe randomized experimental conditions. They do not claim verified compliance.

Assigned perform instructs the participant to initiate the specified imagery after a neutral go signal. Assigned withhold instructs the participant to retain the mapping but not initiate that imagery. Both conditions include an instruction, mapping reminder, cue, delay, go signal, and monitored execution period.

The withhold state is not assumed to be matched to performance in effort or attention. An active nonimagery control is added to evaluate this asymmetry.

3.3 Decomposition of the neural response

Analyses will distinguish four components:

1. **Cue and mapping processing:** activity related to tone identity, reminders, and block context.

2. **Generic instructed implementation or preparation:** activity shared by imagery tasks and the nonimagery active control.
3. **Task-specific content:** independently defined motor-imagery or navigation patterns.
4. **Cross-task generalization:** a perform-minus-withhold direction shared by the two imagery tasks after controlling cue and mapping effects.

A patient may show the fourth component without independently detectable content. Such a result supports recurrent rule-sensitive responding but receives the label **content-unconfirmed ARIS**. The stronger label **imagery-supported ARIS** requires positive motor and navigation content evidence under prespecified criteria.

3.4 Why two imagery tasks remain useful

Motor imagery and spatial navigation depend on partly different neural systems and can fail differently after focal injury. Observed task dissociations support using both [4] [25].

Task generality does not require identical regional activation. The hypothesis is that the signed relation between assigned performance and assigned withholding contains a shared component while each task retains distinguishable content.

3.5 Recurrence does not imply continuity of experience

Recurrence means that a related neural assignment effect is detected on two examinations and transports between them. It does not establish that the patient continuously possesses or experiences the state between visits. It is a repeatability criterion, not a claim about uninterrupted awareness.

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4. Study Design and Clinical Estimand

4.1 Cohorts and sites

The proposed study has four components:

1. an acute/subacute target cohort;
2. a responsive brain-injured control cohort;
3. a healthy calibration and control cohort;
4. a separate prolonged-disorders-of-consciousness transport registry.

Ten sites are proposed. Six development sites will be used for calibration of the prognostic pipeline and internal leave-one-site-out evaluation. Four external sites will remain locked until all estimator, model, and threshold choices are frozen.

The provisional enrollment targets are:

- 600 target-cohort participants;
- 100 responsive brain-injured controls;
- 120 healthy controls;
- up to 120 participants in the prolonged-disorders transport registry.

These numbers are not final. Stage 0 paired-AUC and site-heterogeneity simulations must determine whether they are sufficient.

4.2 Narrowed primary population

The primary cohort is restricted to adults enrolled 10–14 days after acute acquired brain injury. This avoids pooling early and 18-month post-injury patients in one confirmatory estimand.

Inclusion criteria

Participants must:

- be at least 18 years old;
- have traumatic, hypoxic-ischemic, hemorrhagic, ischemic, or other acute acquired brain injury with a documented injury date;
- be examined between post-injury days 10 and 14;
- meet behavioral criteria for UWS or MCS– on serial standardized assessment;
- not meet the study-defined reliable command-following endpoint;
- be sufficiently stable for at least EEG assessment;
- have a legally authorized representative able to provide permission.

For the primary multimodal estimand, sufficient EEG and MRI data must be available by day 28.

Exclusion criteria

Exclusions include:

- pre-injury neurological disease making the outcome uninterpretable;
- uncontrolled status epilepticus;
- physiological instability incompatible with recording;
- research sedation required solely to tolerate testing;
- uncorrectable bilateral auditory-pathway impairment incompatible with the primary spoken task;
- MRI contraindication for the primary multimodal cohort.

Participants unable to undergo MRI may enter an EEG-only observational registry. They will not be classified as ARIS negative.

4.3 Separate prolonged-disorders registry

Patients first enrolled more than 90 days after injury will not be pooled with the acute/subacute confirmatory cohort. They may enter a separately analyzed transport registry. That registry will estimate feasibility and calibration in prolonged disorders of consciousness but cannot confirm the primary day-28-to-day-180 hypothesis.

4.4 Assessment schedule

Primary neural sessions are anchored to injury date rather than enrollment date.

Post-injury time	CRS-R and FOUR	EEG	MRI	Other measurements
Days 10–14	Two CRS-R examinations; FOUR	Rest, tasks, auditory checks	Structural, diffusion, resting and task MRI if stable	Medication, pupil, autonomic and sleep measures
Day 21 ± 2 days	Two CRS-R examinations; FOUR	Rest and tasks	Rest and tasks if safe	Same
Day 28 ± 3 days	Two CRS-R examinations; FOUR	Rest and tasks	Rest and tasks if safe	Same
Day 60 ± 5 days	Two CRS-R examinations; FOUR	Rest and tasks	MRI if safe	Same
Day 90 ± 7 days	Two CRS-R examinations; FOUR	Rest and tasks	MRI if safe	Same
Day 180 ± 14 days	Two CRS-R examinations; FOUR	Optional	Optional	Functional outcome and treatment history
First qualifying command response	Confirmatory examination within 24 hours to 7 days	Extra recording if safe	Extra recording if safe	Medication and arousal review

The primary ARIS score uses only the three early neural sessions. A session obtained after a participant has met reliable command-following cannot contribute to a prebehavioral score.

4.5 Primary risk set

At the day-28 landmark, participants must be:

- alive;
- still under observation;
- not previously meeting reliable command-following;
- technically evaluable under the primary multimodal rules.

This population defines the primary complete-multimodal estimand.

4.6 Primary outcome

The primary outcome is the study-defined emergence of **reliable bedside command-following by post-injury day 180**.

A qualifying examination requires a CRS-R Auditory Function Scale score of 4, corresponding to consistent movement to command under the standard administration, with central review of item-level scoring. The endpoint requires:

- two qualifying examinations;
- different trained examiners;
- separation of at least 24 hours and no more than 7 days;
- no intervening clinical event that invalidates comparison;
- central adjudication blinded to research neural scores.

The two examinations need not use an identical motor effector, because requiring the same movement could penalize focal injury or fluctuating motor access. At least one command must be repeated across the two examinations, and command wording must be documented.

This is a study-defined reliability rule, not a direct quotation from a guideline.

4.7 Aphasia, hearing, and sensory adjudication

Participants with suspected aphasia or uncertain receptive language are not automatically excluded if the auditory pathway is intact. Their status will be documented through:

- pre-injury language history;
- injury location;
- bedside auditory and visual responses;
- brainstem auditory evoked responses where feasible;
- cortical auditory responses;
- a separate spoken-language versus acoustically matched control paradigm;
- alternative visual-command assessment as a secondary outcome.

The primary endpoint remains the standard auditory CRS-R item. Alternative command routes are secondary and cannot be substituted after outcomes are known.

4.8 Death, withdrawal, and observed-care interpretation

For the binary day-180 prediction, a participant who dies before command-following is coded as not having the observed event. Competing-risk analyses will separately distinguish death from survival without command-following.

This outcome combines biological condition with opportunity for observation. It must not be described as the patient's latent ability to recover under unlimited survival and uniform care.

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5. Task and Control Paradigms

5.1 Randomization unit

The cue-to-instruction mapping is assigned at the **mini-block level**. A site-independent randomization service generates a schedule from a finite admissible set subject to:

- equal numbers of the two mappings;
- equal cue counts under each meaning;
- restricted run-order balance;
- limits on mapping repetitions;
- balance of preceding-trial conditions;
- prespecified task and modality order constraints.

Randomization inference will reproduce this exact mechanism. Trial labels will not be treated as freely exchangeable.

5.2 Imagery trial structure

Each imagery task follows the same timing:

1. **Mapping reminder:** spoken instruction identifies the current cue mapping.
2. **Cue:** one of two tones is presented for 500 ms.
3. **Delay:** 2.5 seconds.
4. **Neutral go signal:** identical on all trials.
5. **Execution interval:** 10 seconds.
6. **Intertrial interval:** jittered 4–8 seconds.

Cue-period and reminder-period activity are excluded from the primary execution trajectory and modeled separately.

5.3 Motor-imagery task

On assigned-perform trials, the participant is asked to imagine repeatedly squeezing the right hand, emphasizing kinesthetic sensation and avoiding overt movement. On assigned-withhold trials, the participant is asked to retain the current rule but not initiate hand imagery.

Bilateral forearm electromyography is recorded. Overt movement does not automatically invalidate the entire participant, but contaminated trials are flagged under a frozen rule. An ARIS interpretation cannot rely on a direction reproduced by electromyographic features.

5.4 Spatial-navigation task

On assigned-perform trials, the participant is asked to imagine moving through a familiar environment from a first-person route perspective. On assigned-withhold trials, the participant retains the rule without initiating navigation imagery.

The environment may be selected before injury history is unavailable; standardized familiar-location options are therefore offered. Content heterogeneity is expected and modeled. A failure to express the frozen navigation template may reflect memory or lesion factors rather than absence of rule sensitivity.

5.5 Active nonimagery control

The active control uses the same mapping reminder, arbitrary cues, delay, go signal, execution duration, and reversal schedule. On assigned-perform trials, the participant is instructed to count a sequence of neutral auditory clicks silently. On assigned-withhold trials, the participant retains the mapping and listens without counting.

The control is intended to match:

- rule retrieval;
- delay maintenance;
- anticipation;
- go-cue processing;
- sustained task engagement;
- a nonimagery internal operation.

It cannot be assumed to match subjective effort in noncommunicative patients. The active control therefore constrains, rather than eliminates, generic preparation and effort explanations.

If the imagery ARIS direction is noninferior to and collinear with the auditory-counting direction, the result will be interpreted as generic instructed implementation rather than imagery-specific control.

5.6 Passive replay

Passive replay presents the same sounds and timing without a perform-versus-withhold rule. Pseudo-labels from a matched active schedule are assigned offline. The passive run evaluates whether temporal structure, cue acoustics, drift, or shared event responses generate an apparent connected direction.

Passive replay is a negative control, not an exact randomization test.

5.7 Independent content localizers

Healthy and responsive brain-injured controls complete separate localizer runs used to freeze:

- a motor-imagery content template;
- a navigation-content template;
- an auditory-counting template.

The content templates are not learned from target-patient outcomes. Target participants receive content scores by projection onto these frozen templates.

To qualify as imagery-supported ARIS:

- motor-imagery perform trials must express the motor template beyond assigned-withhold and active-control trials;
- navigation perform trials must express the navigation template beyond assigned-withhold and active-control trials;
- evidence must recur or generalize under a prespecified criterion;
- neither result may be explained by motion or electromyography.

Content support is not required for the primary prognostic score because imposing it might make an already demanding test unusably insensitive. Its absence limits interpretation.

5.8 Trial counts

Provisional targets are:

- 48 imagery trials per task per EEG session;
- 32 imagery trials per task per fMRI session;
- 32 active-control trials per modality;
- one passive-replay run per modality at the first and day-28 sessions.

Trials are grouped into mini-blocks and runs. Cross-fitting leaves out whole mini-blocks or runs, never arbitrarily interleaved trials.

6. An Identifiable Common Representation

6.1 Defect in the previous formulation

The former estimator compared vectors after different modality-specific orthogonal transformations. Although one orthogonal matrix preserves cosine when applied to both vectors,

$$\rho(\mathbf{Oa}, \mathbf{Ob}) = \rho(\mathbf{a}, \mathbf{b}),$$

that identity does not apply to two different maps:

$$\rho(\mathbf{O}_E\mathbf{a}, \mathbf{O}_F\mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{O}_E^\top \mathbf{O}_F \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}.$$

The relative rotation $\mathbf{O}_E^\top \mathbf{O}_F$ can create or reverse alignment. Condition-blind estimation does not remove sign and rotational indeterminacy. The previous mathematical justification was therefore incorrect.

The revised method prohibits modality-specific patient-level rotations.

6.2 External reference space

Let the common reference contain P anatomically named parcels and L prespecified task-phase basis functions. Its dimension is

$$H = PL.$$

Each coordinate has a fixed interpretation:

$$(\text{parcel identity}, \text{phase-basis identity}).$$

Coordinate order is frozen. Signs are tied to signed deviation from a fixed pre-event baseline in an fMRI-anchored reference. No coordinate may be permuted or sign-flipped by patient-specific optimization.

The reference is learned from an independent calibration cohort using:

- simultaneous or closely paired EEG-fMRI rest;
- neutral auditory and somatosensory localizers;
- no target-patient outcome;
- no imagery perform-versus-withhold labels.

Shared event timing may help estimate the encoder. Because shared timing can also encode generic sensory physiology, the calibration battery includes rest and multiple neutral localizers, and Stage 0 explicitly tests whether timing alone creates ARIS-like geometry.

6.3 fMRI reference encoder

For fMRI, parcel-wise finite-impulse-response estimates are projected directly into the fixed parcel-by-phase space. Let

$$\mathbf{x}_n^F \in \mathbb{R}^{p_F}$$

denote fMRI features for calibration observation n. The fixed fMRI operator $\mathbf{A}_F$ consists of anatomical parcellation, lesion-aware coverage rules, deconvolution, and phase-basis projection:

$$\mathbf{y}_n = \mathbf{A}_F \mathbf{x}_n^F \in \mathbb{R}^H.$$

The orientation of $\mathbf{y}_n$ defines the external target.

6.4 EEG-to-reference encoder

Let

$$\mathbf{x}_n^E \in \mathbb{R}^{p_E}$$

contain source-resolved EEG history, frequency-resolved envelopes, and evoked features for calibration observation n. The EEG encoder is defined by the strictly regularized objective

$$\widehat{\mathbf{A}}_E = \arg\min_{\mathbf{A}} \sum_{n=1}^{N_{\mathrm{cal}}} \left\| \mathbf{W}_n^{1/2} \left(\mathbf{y}_n - \mathbf{A}\mathbf{x}_n^E \right) \right\|_2^2 + \lambda \|\mathbf{A}\|_F^2,$$

where:

- $\mathbf{y}_n$ is the fixed fMRI-anchored target;
- $\mathbf{W}_n$ weights observed and reliable coordinates;
- $\lambda > 0$ is selected entirely within calibration data;
- $\|\cdot\|_F$ is the Frobenius norm.

For fixed $\lambda > 0$, the ridge objective has a unique solution under the specified design. The target coordinate signs and order determine the EEG orientation. No Procrustes rotation is subsequently fit to target-patient contrasts.

6.5 Transport to a new participant

For a new participant, the permitted operations are:

- structural registration to the fixed atlas;
- lesion-aware forward modeling;
- condition-blind baseline centering;
- positive coordinate scaling using pre-cue or resting variance;
- masking of coordinates failing predetermined coverage rules.

The prohibited operations are:

- patient-specific rotations;
- reflections or sign changes;
- coordinate permutations;
- alignment optimized using perform-versus-withhold labels;
- mapping chosen to maximize cross-task or cross-modal agreement;
- adaptation using future command-following outcomes.

Positive scaling cannot reverse direction. Anatomical registration may still introduce error, particularly in severely distorted brains, but it cannot arbitrarily rotate one modality toward another in latent space.

6.6 New-site transport

Site calibration uses phantoms, healthy controls, and the neutral calibration battery. Site-specific operations are limited to frozen nuisance correction and positive scale normalization. A site cannot fit a new relative orientation after inspecting patient task effects.

If the external encoder fails predefined calibration-reliability thresholds at a site, that site's primary ARIS analysis is technically invalid until recalibration using outcome-blind data is completed. Recalibrated data cannot be inserted into the original confirmatory analysis after external outcomes are viewed.

6.7 Remaining limitation of the anchor

Algorithmic identifiability does not establish biological equivalence. EEG and fMRI measure different processes and timescales. An EEG encoder trained to predict fMRI-like network states may emphasize vascularly coupled, event-locked, or high-signal processes and may miss valid electrophysiological control patterns. A fixed orientation can therefore be identifiable but scientifically inadequate.

That concern is not resolved by definition. It is addressed through Stage 0 calibration, hemodynamic-distortion simulations, control cohorts, and explicit acknowledgment that failure of the external mapping invalidates the multimodal estimator rather than proving absent patient capacity.

7. Formal Estimator

7.1 Indices and notation

Symbol	Meaning
i	Participant.
q	Task, $q \in \{\mathrm{MI}, \mathrm{SN}\}$.
m	Modality, $m \in \{\mathrm{E}, \mathrm{F}\}$.
s	Early session, $s \in \{14, 21, 28\}$, denoting planned post-injury day.
b	Mini-block.
r	Trial within mini-block.
u_{iqmsbr}	Physical cue identity, -1 or $+1$.
v_{iqmsb}	Mini-block mapping, -1 or $+1$.
c_{iqmsbr}	Assigned instruction, $c = uv$; $+1$ is assigned perform.
$\mathbf{z}_{iqmsbr}$	Trial representation in the fixed H-dimensional reference.
a	Outer cross-fitting superfold.

Symbol	Meaning
$\mathbf{d}^{(a)}_{iqms}$	Held-out assignment contrast for one task-by-modality cell.
$\mathbf{V}^{(a)}_{iqms}$	Covariance estimate for that contrast.
$\mathbf{r}^{(a)}_{iqms}$	Regularized directional vector.
W_{is}	Within-session connected-coherence statistic.
X_{ist}	Cross-session transport statistic for sessions s, t.
R_{ist}	Recurrent score for session pair s, t.
T_i	Raw ARIS statistic, the best recurrent pair after null-calibrated selection.
p^{rand}_i	Design-based assignment probability.
G_i	Development-standardized continuous ARIS predictor.
A_i	Binary ARIS-positive indicator.

7.2 Assignment-effect regression

After conversion to the fixed common space, each coordinate is modeled within task, modality, session, and outer fold. A simplified coordinate-wise model is

$$z_{br} = \alpha_b + \beta_u u_{br} + \beta_c(u_{br} v_b) + \boldsymbol{\theta}^\top \mathbf{h}_{br} + \varepsilon_{br},$$

where:

- α_b is a mini-block intercept;
- β_u captures physical cue identity;
- β_c captures the cue-by-mapping assignment effect;
- $\mathbf{h}_{br}$ contains frozen trial-history and nuisance terms;
- ε_{br} is residual variation.

The vector contrast is

$$\mathbf{d} = 2\widehat{\boldsymbol{\beta}}_c.$$

In a perfectly balanced design, the interaction is orthogonal to cue and mapping main effects. Artifact rejection, missing trials, and unequal precision can destroy that orthogonality. The implemented regression therefore carries the full covariance matrix forward and does not describe post-rejection estimates as automatically orthogonal.

The coefficient identifies an effect of assignment under the randomization assumptions. It does not identify conscious imagery.

7.3 Block-level cross-fitting

Mini-blocks are divided into two or more balanced superfolds before labels and outcomes are examined. Each superfold contains whole mini-blocks or runs. Trial-level random folds are prohibited because temporal autocorrelation and block reminders make adjacent trials dependent.

For outer fold a, nuisance models and any label-adaptive operations are fit without the held-out mini-blocks. The external modality encoder is already frozen and is not re-estimated.

7.4 Low-norm and uncertain directions

Ordinary cosine can become unstable when a contrast norm is small. For each cell, define a directional signal-to-noise quantity

$$Q^{(a)}_{iqms} = \frac{\|\mathbf{d}^{(a)}_{iqms}\|_2}{\sqrt{\mathrm{tr}(\mathbf{V}^{(a)}_{iqms}) + \kappa^2_{qm}}},$$

where $\kappa_{qm} > 0$ is a task-and-modality-specific stabilization constant frozen from independent calibration and control data.

The regularized vector is

$$\mathbf{r}^{(a)}_{iqms} = \frac{\mathbf{d}^{(a)}_{iqms}}{\sqrt{\|\mathbf{d}^{(a)}_{iqms}\|^2_2 + \mathrm{tr}(\mathbf{V}^{(a)}_{iqms}) + \kappa^2_{qm}}}.$$

Its norm is below 1. A weak contrast approaches the zero vector rather than becoming an unstable unit direction.

A binary ARIS classification additionally requires

$$Q^{(a)}_{iqms} \geq Q_{\min}$$

for every task-by-modality cell in both sessions of the selected pair. $Q_{\min}$ is frozen from Stage 0. A cell below this threshold is reported as directionally weak, not technically missing. It prevents weak but accidentally aligned vectors from receiving a favorable binary classification.

7.5 Complete task-by-modality graph

At each session, there are four cell types:

$$\mathcal{H} = \{(\mathrm{MI}, \mathrm{E}), (\mathrm{MI}, \mathrm{F}), (\mathrm{SN}, \mathrm{E}), (\mathrm{SN}, \mathrm{F})\}.$$

The primary graph contains **all six unordered edges** among these four cells. It therefore includes:

- MI-EEG versus MI-fMRI;
- SN-EEG versus SN-fMRI;
- MI-EEG versus SN-EEG;
- MI-fMRI versus SN-fMRI;
- MI-EEG versus SN-fMRI;

- MI-fMRI versus SN-EEG.

The graph is complete and connected. Same-task cross-modal and cross-task within-modal comparisons are mandatory rather than omitted.

For cells $h \neq h'$, the symmetrized held-out projection is

$$C_{ihh's} = \frac{1}{2}\left[\left(\mathbf{r}_{ihs}^{(1)}\right)^{\top}\mathbf{r}_{ih's}^{(2)} + \left(\mathbf{r}_{ihs}^{(2)}\right)^{\top}\mathbf{r}_{ih's}^{(1)}\right].$$

The use of different superfolds reduces shared estimation noise.

7.6 Leave-one-cell common-direction tests

For held-out cell h, estimate the training direction from the other three cells:

$$\mathbf{g}_{i,-h,s}^{(a)} = \frac{\sum_{h' \neq h} w_{ih's}^{(a)}\mathbf{r}_{ih's}^{(a)}}{\left\|\sum_{h' \neq h} w_{ih's}^{(a)}\mathbf{r}_{ih's}^{(a)}\right\|_2},$$

provided the denominator exceeds an externally frozen resultant threshold. Otherwise the corresponding validation score is set to zero and cannot support positivity.

The held-out score is

$$L_{ihs} = \frac{1}{2}\left[\left(\mathbf{g}_{i,-h,s}^{(1)}\right)^{\top}\mathbf{r}_{ihs}^{(2)} + \left(\mathbf{g}_{i,-h,s}^{(2)}\right)^{\top}\mathbf{r}_{ihs}^{(1)}\right].$$

All four L_{ihs} values must be favorable.

7.7 Leave-one-task and leave-one-modality tests

For leave-one-task testing, a direction is estimated from both modalities of MI and tested separately in both SN modalities; the reverse direction is also tested. For leave-one-modality testing, a direction is estimated from both tasks in EEG and tested separately in both fMRI tasks; the reverse direction is also tested.

Let the resulting projection sets be $\mathcal{L}_{is}^{\text{task}}$ and $\mathcal{L}_{is}^{\text{mod}}$. These tests ensure that generalization is not an artifact of pooling all cells before validation.

7.8 Within-session statistic

The within-session connected score is the minimum of all required components:

$$W_{is} = \min\left(\{C_{ihh's} : h < h'\}, \{L_{ihs} : h \in \mathcal{H}\}, \mathcal{L}_{is}^{\text{task}}, \mathcal{L}_{is}^{\text{mod}}\right).$$

A favorable average cannot conceal one opposed task-by-modality cell because the weakest required component determines W_{is}.

7.9 Cross-session transport

For sessions $s < t$, a consensus direction estimated in session s is tested in every cell of session t, and vice versa. Same-cell recurrence and all cross-cell projections are also included.

The cross-session statistic is

$$X_{ist} = \min\left(\mathcal{X}_{s\to t}, \mathcal{X}_{t\to s}, \mathcal{X}_{\text{same cell}}, \mathcal{X}_{\text{leave task}}, \mathcal{X}_{\text{leave modality}}\right).$$

Every component uses independent superfolds across the two sessions. The detailed ordered list of projections will be generated automatically from a frozen graph specification and included in the executable preregistration.

7.10 Recurrent session-pair statistic

For session pair s, t,

$$R_{ist} = \min(W_{is}, W_{it}, X_{ist}).$$

The raw patient score is

$$T_i = \max_{(s,t)\in\{(14,21),(14,28),(21,28)\}} R_{ist}.$$

Thus, a participant must have one pair of individually coherent sessions with successful bidirectional transport. The maximum permits one of the three planned sessions to fail because of biological fluctuation or measurement noise. Selection among the three pairs is included inside the design-based randomization distribution, so the repeated opportunity is not ignored.

7.11 Rejection of the checkerboard alternative

Consider the pattern

$$\mathbf{d}_{\text{MI,E}} = \mathbf{g}, \qquad \mathbf{d}_{\text{SN,F}} = \mathbf{g},$$

$$\mathbf{d}_{\text{MI,F}} = -\mathbf{g}, \qquad \mathbf{d}_{\text{SN,E}} = -\mathbf{g}.$$

A score using only cross-task, cross-modal diagonal pairs could treat this as perfect alignment. The revised complete graph contains same-task cross-modal and cross-task within-modal edges, each of which has negative projection under this pattern. The within-session minimum is therefore negative. Leave-one-cell consensus also fails because the held-out direction is opposed to at least part of the training set.

Checkerboard rejection must additionally be demonstrated in finite-sample simulation before registration.

7.12 Common-direction reporting

After a participant passes the connected tests, a descriptive common direction may be estimated from all valid cells in the selected session pair:

$$\widehat{\mathbf{g}}_i = \frac{\sum_{h,s\in\mathcal{S}_i^\star} w_{ihs}\mathbf{r}_{ihs}}{\left\|\sum_{h,s\in\mathcal{S}_i^\star} w_{ihs}\mathbf{r}_{ihs}\right\|_2}.$$

This direction is reported for interpretation and content decomposition. It is not used to retroactively redefine the primary score.

7.13 Geometric identity and its limited role

For J unit vectors and all unordered pairs,

$$\left\|\sum_{j=1}^{J}\mathbf{e}_j\right\|_2^2 = J + 2\sum_{j<k}\mathbf{e}_j^\top\mathbf{e}_k.$$

The identity correctly characterizes the mean over a complete unweighted graph. The primary statistic is instead a minimum over regularized held-out projections plus transport tests. The identity is therefore only a descriptive visualization of concentration and is not presented as a derivation of T_i.

7.14 Common-direction signal model

For simulation, a planted common-direction model is

$$\mathbf{d}_{ihs} = \lambda_{ihs}\mathbf{g}_i + \boldsymbol{\epsilon}_{ihs}, \qquad \lambda_{ihs} \geq 0.$$

Approximate expected cosine formulas require more than isotropic mean-zero error. Simulations will vary:

- cross-cell error covariance;
- projection of error onto $\mathbf{g}_i$;
- concentration of random norms;
- modality-dependent noise;
- session autocorrelation;
- signal-to-noise ratio;
- lesion-related coordinate loss.

No analytic ratio approximation will be used for primary inference.

7.15 Continuous prognostic predictor

The external prognostic model uses

$$G_i = \frac{T_i - \mu_T^{\mathrm{dev}}}{\sigma_T^{\mathrm{dev}}},$$

where the mean and standard deviation are estimated in the development cohort without external outcomes and frozen before external processing.

G_i is not assumed to be normally distributed. The maximum over session pairs, minimum over components, variable precision, and eligibility rules make normality implausible. “Per standard deviation” is only a frozen scaling convention.

7.16 Binary ARIS classification

A participant is ARIS positive only if all of the following hold:

1. $p_i^{\mathrm{rand}} \leq 0.025$;
2. $T_i \geq \delta_{\min}$, with $\delta_{\min}$ frozen in Stage 0;

3. every cell in the selected session pair passes $Q_{\min}$;
4. all technical data-quality requirements are met;
5. passive replay and motion-control failure rules are not triggered.

The continuous G_i is primary for prognosis. Binary status is used for prevalence, recurrence, and landmark timing.

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8. Randomization Inference and Negative Controls

8.1 Exact assignment null

The primary inferential null is the sharp assignment null that the observed neural data would be unchanged under another admissible mini-block cue-mapping schedule.

Let Ω_{iqms} be the finite set of schedules allowed by the actual randomization restrictions. For each realization:

1. draw a schedule from Ω_{iqms}, or enumerate Ω if feasible;
2. reconstruct $c = uv$;
3. refit every label-adaptive nuisance or contrast model;
4. reapply cell-reliability rules;
5. rebuild the complete graph;
6. recompute every candidate session pair;
7. select the maximum recurrent pair exactly as in observed data;
8. return $T_i^{(b)}$.

The empirical probability is

$$p_i^{\text{rand}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}(T_i^{(b)} \geq T_i)}{B+1}.$$

Ties count against significance.

8.2 Number of randomizations

The protocol specifies

$$B = 39{,}999$$

for the primary patient-level analysis unless all schedules are enumerated. This provides resolution well below 0.025. A binomial Monte Carlo confidence interval for the estimated tail probability will be reported.

There is no normal-theory $Z > 1.96$ requirement. Null standard deviations are not used, so a zero or nearly zero null variance cannot generate an undefined or inflated z-score. If all randomized statistics are identical, the probability is 1 unless the observed statistic is uniquely larger under enumeration, which should not occur under the fixed procedure.

8.3 Why one exact test replaces four nominal null probabilities

The earlier procedure took a minimum across label, cue, temporal-shift, and phase surrogate z-scores. That construction was not a calibrated global probability, and several components lacked exchangeability.

The revised primary probability is tied only to the actual randomized allocation. The other analyses are explicitly negative controls or model checks. Requiring an exact assignment test does not make phase-randomized or temporally shifted values exact tests.

8.4 Cue-identity analysis

A separate statistic replaces $c = uv$ with physical cue identity u. It asks whether the same physical sound generates a recurrent graph. Because auditory cue responses may be real, this is not expected always to be zero.

The rule-sensitive interpretation fails if:

- the assignment statistic does not exceed the cue-identity statistic by the frozen margin δ_{cue}; or
- the delayed execution direction remains fixed to one physical cue after mapping reversal.

8.5 Temporal-shift distribution

Execution labels are circularly shifted by offsets exceeding a full trial and avoiding exact block-period aliases. The procedure tests whether slow drift or periodic timing can reproduce the statistic.

Circular shifts are not assumed exchangeable. Their tail proportions are called **temporal-shift exceedance rates**, not p-values.

8.6 Phase-randomized distribution

Fourier or wavelet phases are randomized within continuous runs under a frozen algorithm. Shared phase increments may preserve some cross-spectral structure, but they do not establish exchangeability with the event sequence. These surrogates are used to diagnose spectral and temporal fragility.

Nonstationary segments, motion bursts, and state transitions are separately marked. Stage 0 must show how the surrogate behaves under realistic nonstationarity before it is interpreted.

8.7 Passive and active controls

The passive-replay positivity rate is estimated with confidence bounds in healthy and responsive controls. The mandatory Stage 0 gate requires the one-sided upper 95% confidence bound for passive ARIS positivity to remain below 0.05.

The active nonimagery control is not expected to be null. It is used to decompose generic instructed implementation from imagery-related generalization.

8.8 Repeated landmark opportunities

The day-28 analysis is primary and uses the fixed three-session recurrence rule. Later day-60 and day-90 classifications are secondary. Their predictive tests use prespecified one-sided alpha levels of 0.015 and 0.010, respectively. They cannot replace a failed day-28 primary result.

9. Neural Acquisition and Measurement Controls

9.1 EEG acquisition

High-density EEG will be recorded with harmonized hardware targets and synchronized electrooculography, electromyography, electrocardiography, event markers, and video where permitted.

Preprocessing includes:

- bad-channel detection;
- line-noise attenuation;
- robust rereferencing;
- muscle and movement screening;
- lesion-aware source modeling;
- frequency-resolved source features;
- trial and mini-block quality metrics;
- blinded manual review of automatically flagged segments.

Quality reviewers remain blind to condition labels and outcomes.

9.2 Lesion-aware source reconstruction

Severe injury can alter anatomy, conductivity, skull integrity, and electrode-to-cortex correspondence. Generic atlas source models are therefore insufficient for the primary analysis.

Where feasible, EEG forward models will incorporate:

- individual structural MRI;
- lesion masks;
- skull defects or cranioplasty;
- electrode digitization;
- uncertainty in tissue conductivity;
- alternative plausible head models.

Source-leakage sensitivity analyses will vary inverse solutions and parcel resolution. Phase-lagged connectivity may reduce some zero-lag contamination but is not treated as eliminating volume conduction, source leakage, or common input.

If individualized modeling is impossible, that session can remain in scalp-level secondary analysis but is not automatically eligible for primary source-space ARIS.

9.3 fMRI acquisition

MRI includes:

- structural lesion-sensitive imaging;
- diffusion MRI;
- resting fMRI;
- both imagery tasks;

- active control;
- passive replay where feasible;
- perfusion or cerebrovascular-reactivity assessment where feasible.

Task responses are estimated with finite-impulse-response models. A canonical hemodynamic response is not imposed as the sole model.

9.4 Neurovascular uncertainty

Finite-impulse-response modeling does not solve altered neurovascular coupling. Injury, vascular damage, carbon-dioxide variation, medication, regional hypoperfusion, and delayed hemodynamics can change fMRI without corresponding neural change.

The protocol therefore includes:

- regional response-latency estimation;
- perfusion or vascular-reactivity measurements where feasible;
- coverage and susceptibility masks;
- sensitivity to delayed and broadened hemodynamic responses;
- simulation of plausible regional distortions;
- comparison with EEG in the fixed external space.

A cross-modal mismatch may be vascular rather than cognitive. It is a failure of the multimodal test, not evidence that the participant lacks rule sensitivity.

9.5 Diffusion MRI

Diffusion measures are described as model-dependent proxies for microstructural organization and tractography continuity. They do not directly establish that a pathway is physically intact, synaptically effective, or functionally communicating.

The protocol will quantify uncertainty across tractography settings and will not use “route available” as an unqualified synonym for a diffusion streamline.

9.6 Auditory and language measures

Because the task is instruction dependent, every primary session includes:

- ear-specific sound-level verification;
- bedside auditory examination;
- auditory evoked responses where feasible;
- a cortical deviance or sound-discrimination response;
- a spoken-sentence versus acoustically matched control paradigm;
- documentation of language history and dominant language;
- documentation of aphasia-relevant lesions.

These checks do not prove comprehension. They localize some false-negative pathways and qualify positive interpretations.

9.7 Arousal, sleep, and medication

At each session, the study records:

- pupil diameter and variability;
- eye opening;
- sleep stage and recent sleep;
- heart rate and variability;
- oxygenation and ventilation;
- blood pressure where available;
- temperature;
- sedatives;
- antiseizure agents;
- analgesics;
- stimulants and dopaminergic agents;
- infection and metabolic disturbance;
- seizure activity;
- clinical neuromodulation.

Sleep-pattern assessment and clinical EEG are consistent with multimodal diagnostic guidance [3]. Clinically necessary medication is never withheld for research. Research testing is postponed during deep sedation where possible.

9.8 Motion and muscle controls

Motion features are analyzed as their own candidate decoder. A neural interpretation fails if motion alone:

- classifies assigned perform versus withhold;
- follows cue reversal;
- recurs across sessions;
- reproduces the connected task-by-modality graph.

Motor-task analyses are repeated after excluding all trials with electromyographic concern. Navigation and active-control results must not depend on motor contamination.

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10. Prognostic Modeling and External Validation

10.1 Predictor sets

The complete baseline model $\mathcal{B}_0$ contains prespecified groups:

1. age, represented by a frozen nonlinear basis;
2. sex, recorded but interpreted descriptively;
3. etiology;
4. baseline UWS versus MCS−;

5. CRS-R total and relevant item scores;
6. FOUR score;
7. sedation and arousal burden;
8. structural lesion measures;
9. resting EEG measures;
10. resting-fMRI measures;
11. diffusion proxies;
12. acquisition count, valid mini-block count, motion burden, modality completeness, and session timing;
13. all four isolated task-by-modality instruction-sensitivity scores.

Including all four isolated task scores avoids selecting a favorable “best” task after outcome inspection. A separate one-task comparator will still be reported, with its identity selected inside development resampling and frozen before external validation.

The full model $\mathcal{B}_1$ adds continuous ARIS G_i.

10.2 Dimensionality control

The baseline is deliberately rich but cannot be fit as an unrestricted model. Continuous predictors are standardized using development values. Prespecified nonlinear bases are limited to age, CRS-R total, and time-related variables. No unplanned interactions are permitted in the confirmatory model.

Coefficients are estimated using ridge-penalized hierarchical logistic regression. Group-specific penalties are selected through nested leave-one-development-site-out validation. The entire penalty-selection process is repeated inside development bootstrap analyses. External outcomes are never used for:

- feature selection;
- penalty selection;
- single-task selection;
- structural-score construction;
- nonlinear transformation choice.

Selection stability and effective degrees of freedom will be reported.

10.3 Site model and unseen-site prediction

Let η_j denote a development-site intercept:

$$\eta_j \sim \mathcal{N}(0, \tau^2_{\text{site}}).$$

The conditional model is

$$\Pr(R_i = 1 \mid \mathbf{X}_i, G_i, \eta_{j(i)}) = \operatorname{logit}^{-1}\left(\alpha + \boldsymbol{\gamma}^\top \mathbf{X}_i + \beta_G G_i + \eta_{j(i)}\right).$$

For a new external site, the primary probability is the site-marginal prediction

$$\hat{p}_i = \int \operatorname{logit}^{-1}\left(\alpha + \boldsymbol{\gamma}^\top \mathbf{X}_i + \beta_G G_i + \eta\right) \phi(\eta; 0, \hat{\tau}^2_{\text{site}})\, d\eta.$$

The integral is computed using frozen numerical quadrature.

Setting $\eta = 0$ would instead give a conditional prediction for a hypothetical average-effect site. It is not equal to the marginal probability because the logistic function is nonlinear. Conditional $\eta = 0$ predictions will be reported only as a sensitivity analysis.

Six development sites may provide weak information about τ_{site}. The variance receives a regularizing prior or penalty specified before outcome access, and fixed-site and site-stratified alternatives are evaluated in Stage 0 simulations.

10.4 Primary discrimination criteria

For predicted score S,

$$\text{AUC} = \Pr(S_{i_1} > S_{i_0}),$$

with half credit for ties.

Incremental discrimination is

$$\Delta\text{AUC} = \text{AUC}(\mathcal{B}_1) - \text{AUC}(\mathcal{B}_0).$$

Confirmatory success requires:

$$\text{LCB}_{97.5\%}\{\text{AUC}(\mathcal{B}_1)\} > 0.80$$

and

$$\text{LCB}_{97.5\%}\{\Delta\text{AUC}\} > 0.10.$$

Because this is an intersection-union claim, both one-sided tests must pass. If either fails, the study does not confirm the primary hypothesis.

10.5 Decision states

Four result states are prespecified.

1. **Confirmed:** both lower confidence bounds exceed their targets and all validity guardrails pass.
2. **Not confirmed:** at least one lower bound does not exceed its target.
3. **Strong evidence below target:** an upper one-sided 97.5% bound is below 0.80 for AUC or below 0.10 for incremental AUC.
4. **Technically invalid:** estimator, external mapping, control sensitivity, or data-integrity gates fail.

"Not confirmed" includes an inconclusive region. It is not automatically evidence of no association.

10.6 Calibration and overall accuracy

The following are mandatory reports:

- calibration intercept;
- calibration slope;

- Brier score;
- sensitivity and specificity at frozen thresholds;
- positive and negative predictive values at observed prevalence;
- decision-curve summaries;
- distribution of predictions by outcome;
- calibration by site.

AUC confirmation does not imply clinically acceptable calibration.

10.7 Site-specific performance

A site-specific AUC is reported only if that site contains at least eight participants with and eight without the outcome. If fewer than three of four external sites meet this requirement, site-level transportability is considered inadequately evaluated even if the pooled AUC is estimable.

The macro-average AUC is calculated only across estimable sites, with the omitted sites and denominators explicitly reported. It cannot be silently defined for a one-class site.

A transportability guardrail requires:

- at least three estimable external sites;
- no estimable site AUC point estimate below 0.65;
- an upper 95% bound for the between-site standard deviation on the logit-AUC scale below the Stage 0-frozen tolerance, provisionally 0.35.

The exact tolerance remains conditional on simulation.

10.8 External uncertainty

The primary external estimand is performance in the mixture of the four participating external sites, not every conceivable future hospital.

Confidence intervals will use a site-stratified patient bootstrap with 10,000 replicates, paired resampling for model differences, and propagation of development-model uncertainty through frozen development bootstrap fits. Sites are not naively resampled as though four clusters could support stable superpopulation inference.

Stage 0 must demonstrate acceptable coverage for the chosen interval method under the planned number of sites and outcome events. If coverage is inadequate, a calibrated Bayesian or permutation-based alternative must be frozen before recruitment.

10.9 Confounding and coefficient interpretation

The earlier rule that adjustment should reduce the ARIS logistic coefficient by less than 30% is removed. Logistic coefficients are non-collapsible and may change after adding prognostic variables without confounding.

Robustness will instead be assessed through:

- standardized risk differences;
- externally predicted probabilities;

- paired changes in AUC and Brier score;
- fixed covariate distributions;
- negative controls;
- models including arousal, medication, motion, acquisition quantity, and treatment history.

These analyses remain associational. A causal interpretation would require an explicit treatment and measurement model beyond ordinary adjustment.

10.10 Equal-information sensitivity analysis

ARIS precision can improve merely because a participant tolerates more usable sessions and trials. Tolerance itself may predict recovery.

The primary baseline therefore includes:

- number of acquired and valid mini-blocks;
- modality completeness;
- motion burden;
- session duration;
- interval between modalities;
- timing relative to injury.

A fixed-information sensitivity analysis subsamples every participant to the same number of mini-blocks and the same two-session pattern before recomputing ARIS. If prognostic value disappears under equal information, acquisition opportunity is a plausible explanation.

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11. Dynamic Landmark and Longitudinal Analyses

11.1 Why visit-order shuffling is not used

Visits are biologically ordered, unequally spaced, and subject to informative missingness. Shuffling their order would create impossible histories and would not represent a valid temporal null. The earlier visit-order permutation is abandoned.

11.2 Fixed dynamic landmarks

Analyses are conducted at post-injury days 28, 60, and 90.

At each landmark L, the risk set includes only participants who are:

- alive;
- observed;
- not already meeting reliable command-following;
- evaluable using measurements obtained no later than L.

The model predicts command-following in $(L, 180]$, with death treated as a competing event in time-to-event analyses.

The day-28 landmark is confirmatory. Later landmarks describe updating and cannot rescue it.

11.3 Lead time

For an ARIS-positive participant, T_A is the date of the second session in the first qualifying recurrent pair. For the behavioral endpoint, T_R is the first qualifying CRS-R examination once confirmed by the second examination.

Lead time is

$$L_i = T_{Ri} - T_{Ai}.$$

Because assessments are intermittent, lead time is interval-censored. It will be summarized with bounds and medians rather than assigned spurious day-level precision.

11.4 Time-dependent performance

At each landmark, the study reports:

- time-dependent AUC;
- sensitivity and specificity for command-following by day 180;
- cumulative incidence of command-following;
- cumulative incidence of death;
- median and distribution of lead time;
- performance conditional on survival as a sensitivity estimand.

These analyses directly assess prediction before behavior without conditioning solely on future recoverers.

11.5 Observation and dropout process

Missing visits can depend on illness, transfer, death, inability to tolerate MRI, and emerging recovery. A joint longitudinal-event model will include:

- the ARIS trajectory;
- the command-following event;
- death;
- the visit-observation process;
- time-varying medication and instability.

The joint model is secondary because it depends on stronger assumptions than the fixed day-28 prediction.

11.6 Continuous-time longitudinal model

Unequal intervals make a discrete session-index cross-lag model inappropriate. Let

$$\mathbf{Y}_i(t) = \begin{bmatrix} G_i(t) \\ M_i(t) \end{bmatrix},$$

where $M_i(t)$ is the mandatory network-support index defined below. A continuous-time state-space model is proposed:

$$d\mathbf{Y}_i(t) = \mathbf{A}\left[\mathbf{Y}_i(t) - \boldsymbol{\mu}_i(t)\right] dt + \mathbf{B}\mathbf{W}_i(t)\, dt + \mathbf{L}\, d\mathbf{W}_i^*(t).$$

Here:

- $\mathbf{A}$ contains autoregressive and cross-process drift terms;
- $\boldsymbol{\mu}_i(t)$ contains participant-level trajectories;
- $\mathbf{W}_i(t)$ contains measured treatments and arousal variables;
- $d\mathbf{W}_i^*(t)$ is stochastic innovation;
- elapsed time enters continuously.

The off-diagonal term from M to G is compared with the reverse term. Missing measurements are handled through the observation model rather than pretending that every session lag is equal.

Temporal asymmetry can support ordering. It cannot establish that network restoration causes ARIS.

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12. Mechanistic and Neuroscientific Analyses

12.1 Mandatory network-support index

The mandatory longitudinal index excludes optional TMS. Its components are frozen and directionally specified:

1. long-range alpha participation and modular span;
2. interhemispheric frontoparietal phase-lagged connectivity;
3. thalamocortical diffusion proxy;
4. model-dependent resting-fMRI coupling within the prespecified anterior forebrain–frontoparietal model.

Components are normalized against control distributions and combined with fixed equal weights only after each is oriented according to a prespecified hypothesis. Clustering coefficient and local efficiency are not treated as uniformly favorable because some reported values are higher in UWS than MCS [34].

If one mandatory component is missing, the composite is not silently recomputed from a different set. A formally specified latent-variable missing-component model is used in secondary analysis; the complete-component score remains primary for the mechanistic test.

12.2 Alpha-network hypothesis

Prior EEG evidence supports associations among alpha-network efficiency, long-range organization, behavioral awareness, and some active-neuroimaging responses [2]. The present study predicts that higher long-range alpha organization will precede stronger ARIS.

This is a proposed association. It does not make alpha organization necessary or sufficient.

12.3 Frontoparietal and thalamocortical hypotheses

The protocol predicts that ARIS will be more common when:

- interhemispheric frontoparietal connectivity is relatively preserved;
- thalamocortical diffusion proxies are less disrupted;
- the frozen effective-connectivity model assigns stronger coupling to selected anterior forebrain and cortical paths.

Neuroimaging reviews support considering thalamocortical circuits and differentiated network communication [35]. Model-based anterior forebrain findings provide a specific but preprint-level hypothesis [38].

Effective-connectivity parameters remain conditional on the chosen generative model. They are not described as observed information flow.

12.4 Primary mechanistic decision rule

The primary mechanistic parameter is the standardized continuous-time drift contrast

$$\Delta_{\mathrm{mech}} = A_{G\leftarrow M} - A_{M\leftarrow G},$$

expressed as the difference in expected standardized 30-day change.

Mechanistic support requires:

$$\mathrm{LCB}_{97.5\%}(\Delta_{\mathrm{mech}}) > 0.10.$$

Mechanistic futility is declared if:

$$\mathrm{UCB}_{97.5\%}(\Delta_{\mathrm{mech}}) < 0.10.$$

Values spanning 0.10 are inconclusive. Even a positive result is evidence of temporal ordering under the model, not proof of causation.

12.5 Perturbational substudy

A separately powered, optional TMS–EEG substudy will test network propagation. It does not contribute to the mandatory network index or primary prognostic model.

The substudy requires:

- independent safety review;
- explicit skull-defect, implant, seizure, and medication exclusions;
- active and sham stimulation;
- auditory masking;
- matched scalp somatosensory control;
- blinded condition assessment;
- frozen pulse-artifact removal;
- target-specific and sham-specific propagation measures.

Perturbational approaches have been described as promising for consciousness assessment in severely damaged brains [26]. That source does not validate this TMS implementation or establish safety in the proposed population. Safety and interpretability must be established independently before the substudy opens.

12.6 Mechanistic dissociations

Several outcomes are informative:

- **ARIS positive, network index low:** challenges a broad enabling-network requirement or suggests sparse reorganization.
- **Network index high, ARIS negative:** suggests permissive organization without detected instruction sensitivity, or task failure.
- **Perturbational propagation positive, ARIS negative:** suggests propagation capacity without successful task implementation.
- **ARIS prognostic, mechanism null:** supports ARIS as a biomarker while rejecting the proposed network ordering.
- **Mechanism associated, ARIS nonincremental:** suggests ARIS is a redundant expression of broader network recovery.

The prognostic and mechanistic hypotheses can therefore fail separately.

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13. Missing Data, Selection, Treatment, and Sensitivity Analyses

13.1 Data states

Every task-by-modality cell is assigned one state:

- technically valid;
- technically limited but analyzable;
- directionally weak;
- technically invalid;
- missing.

Directionally weak means that data exist but the contrast does not meet $Q_{\min}$. Technically invalid means that artifact, coverage, randomization, or preprocessing failure prevents interpretation. Neither is silently recoded as a positive or omitted according to outcome.

13.2 Primary complete-multimodal estimand

The confirmatory estimand applies to participants with sufficient EEG and fMRI information for the complete connected graph at the day-28 landmark. This is explicitly an MRI-complete, technically evaluable population.

It may exclude medically unstable or severely injured patients. Performance cannot be generalized to all behaviorally nonresponsive patients without additional evidence.

13.3 Selection analyses

Selection into the primary estimand will be modeled using:

- baseline severity;
- etiology;
- lesion burden;
- physiological instability;
- site;
- MRI contraindication;
- transfer;
- session completion;
- treatment and care limitation.

Sensitivity analyses include inverse-probability weighting and pattern-mixture shifts. These methods cannot recover information lost through unmeasured selection. Complete-case and weighted estimands will be reported separately.

13.4 Baseline covariate missingness

Baseline predictor missingness is handled through a frozen multiple-imputation pipeline developed entirely within development resampling. The external outcome is never used to generate an external predictor.

ARIS itself is not imputed for the confirmatory complete-multimodal analysis. Imputed ARIS analyses are explicitly secondary.

13.5 Death and withdrawal

Secondary analyses include:

- cause-specific hazard models;
- cumulative-incidence models with death as a competing event;
- survivor-conditional estimates;
- best-case and worst-case withdrawal bounds;
- inverse-probability-of-censoring weights;
- a three-category outcome distinguishing command-following, survival without command-following, and death.

No one analysis fully resolves the clinical meaning of death before possible recovery.

13.6 Treatment exposure

Time-varying models record sedatives, antiseizure medication, analgesics, amantadine, other dopaminergic agents, neuromodulation, rehabilitation intensity, infection, seizures, and major complications.

Treatment confounding by indication remains likely. The study predicts outcomes under observed care and does not estimate an untreated natural history.

13.7 Disclosure as an intervention on care

Research findings may alter clinician or family decisions. Disclosure status and subsequent changes in rehabilitation, medication, and life-sustaining treatment will therefore be treated as time-varying events.

No statistical adjustment can fully remove feedback if a biomarker changes the outcome it predicts. The primary analysis will report both the disclosure policy and the proportion of participants whose research findings entered clinical discussion.

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14. Stage 0 Simulation and Calibration Program

14.1 Purpose

Stage 0 is mandatory because the revised estimator is mathematically specified but not empirically validated. No patient recruitment begins until the simulation code, calibration data, and decisions are locked and publicly timestamped.

14.2 Planted-signal scenarios

Simulations will include:

1. one common signed direction in every task-by-modality cell;
2. task-specific directions with no common component;
3. modality-specific directions;
4. session-specific directions;
5. the checkerboard $+\mathbf{g}, -\mathbf{g}$ alternative;
6. one favorable cell and three null cells;
7. generic preparation shared with the active control;
8. physical cue effects without mapping sensitivity;
9. cue-by-block association without imagery content;
10. slow drift and periodic block structure;
11. realistic EEG and fMRI autocorrelation;
12. altered hemodynamic delays and dispersion;
13. low-norm contrasts;
14. correlated cell errors;
15. degenerate calibration covariance;
16. missing cells and unequal session counts;
17. acquisition quality correlated with future outcome;
18. site shifts in prevalence, noise, and calibration.

14.3 Mapping perturbations

The frozen common-space procedure will be challenged by:

- artificial sign flips;

- coordinate permutations;
- relative rotations;
- calibration-sample changes;
- source-localization error;
- lesion-mask perturbation;
- scanner scaling and delay changes.

The implemented pipeline should remain invariant to permitted positive scaling but should fail transparently, rather than appear positive, when prohibited orientation changes are introduced.

14.4 False-positive gate

Under every no-common-direction and checkerboard scenario, the upper 95% simulation confidence bound for the full ARIS-positive rate must be no greater than 0.03 at the nominal 0.025 assignment threshold.

Calibration is assessed for the complete procedure, including:

- mini-block re-randomization;
- cross-fitting;
- reliability thresholds;
- graph minima;
- selection of the best session pair;
- missingness rules.

Nominal component behavior is insufficient.

14.5 Recovery gate

At the prespecified minimally relevant signal-to-noise ratio, the lower 95% simulation confidence bound for recovery of the planted direction must exceed 0.70. Sign recovery, cell-wise projection, selected session pair, and direction similarity are all evaluated.

The signal-to-noise value defining “minimally relevant” must be justified from independent control data before the gate is applied.

14.6 Control-cohort gates

The study proceeds only if:

- healthy-control ARIS positivity is at least 75%, with a one-sided lower 95% confidence bound above 65%;
- responsive brain-injured-control positivity is at least 60%, with a lower bound above 45%;
- passive-replay positivity has a one-sided upper 95% bound below 5%;
- test–retest direction similarity meets a frozen reliability criterion;
- the external EEG-to-reference encoder transports across held-out sites.

Failure in healthy controls invalidates the estimator or task. Failure in responsive injured controls indicates inadequate clinical transportability. Neither can be reinterpreted as evidence that the test detects a subtler capacity.

14.7 Prognostic power simulations

Power calculations must simulate the actual co-primary rule, not AUC alone. They will vary:

- event prevalence from 0.25 to 0.50;
- full-model AUC from 0.70 to 0.88;
- paired incremental AUC from 0.03 to 0.15;
- correlation between baseline and full-model scores;
- site variance;
- external-site event imbalance;
- calibration drift;
- technical invalidity from 0.10 to 0.35;
- treatment-related censoring;
- death and withdrawal;
- development-model uncertainty.

The final recruitment target must provide at least 80% probability of satisfying both lower-confidence-bound criteria under the minimally acceptable alternative chosen before recruitment. If the provisional 600-patient target is insufficient, it will be increased or the study will be declared infeasible before enrollment begins.

14.8 No simulated results are claimed here

This manuscript specifies the simulation program but does not report its outcome. The reviewers' request for completed recovery and false-positive evidence is therefore not treated as resolved. It remains a condition precedent to preregistration.

15. Quantitative Predictions and Falsification Rules

15.1 Predictions

If Stage 0 is passed, the following are prospective predictions rather than findings.

Domain	Prediction
Healthy controls	At least 75% ARIS positive under the full recurrent rule.
Responsive injured controls	At least 60% ARIS positive.
Passive replay	Upper one-sided 95% confidence bound for positivity below 5%.
Checkerboard simulation	Upper 95% bound for false positivity no greater than 3%.
Target cohort	ARIS positive less often than any isolated task decoder.
External full model	Lower one-sided 97.5% AUC bound above 0.80.
Incremental value	Lower one-sided 97.5% paired ΔAUC bound above 0.10.
Calibration	Slope point estimate between 0.8 and 1.2, reported with uncertainty.

Domain	Prediction
Temporal prediction	Day-28 ARIS predicts command-following after day 28.
Mechanistic ordering	Lower one-sided 97.5% bound for Δ_{mech} above 0.10.

Only the formal confidence-bound rules determine confirmation.

15.2 Estimator invalidity

The estimator is invalid if any mandatory Stage 0 or control-cohort gate fails. Patient results obtained with an invalid estimator cannot be interpreted as biological negatives.

15.3 Primary prognostic failure

The primary hypothesis is not confirmed if either lower one-sided 97.5% bound fails to exceed its target. Strong evidence below target is declared if the corresponding upper bound is below target.

15.4 Task-generality failure

Task generality fails if:

- any task-by-modality cell is systematically opposed;
- checkerboard alternatives pass above the allowed rate;
- leave-one-task or leave-one-modality transport fails;
- a single-task model is noninferior to the full ARIS model within a frozen AUC margin of 0.02 and ARIS adds less than 0.10.

The noninferiority margin is evaluated through the paired external confidence interval.

15.5 Rule-sensitive interpretation failure

The rule-sensitive interpretation fails if:

- physical cue identity explains the recurrent direction within δ_{cue};
- assignment effects do not reverse with the mapping;
- passive replay passes above the control limit;
- motion or electromyography reproduces the graph;
- the exact assignment test is not calibrated.

15.6 Imagery interpretation failure

The imagery-supported interpretation fails if independently frozen motor and navigation content criteria are not met. The participant may remain ARIS positive but must be labeled content unconfirmed.

If active auditory counting yields a direction noninferior to the imagery direction within a frozen projection margin of 0.10, the common component is interpreted as generic instructed implementation or preparation rather than imagery-specific control.

15.7 Temporal failure

The early-prediction hypothesis fails if day-28 ARIS does not predict events after the landmark with the specified external criteria. Concurrent ARIS at or after bedside command-following cannot rescue the early-prediction claim.

15.8 Mechanistic failure

The proposed network ordering is futile if the upper confidence bound for the standardized 30-day drift contrast is below 0.10. A prognostic ARIS result can survive mechanistic failure, but the frontoparietal–thalamic ordering claim cannot.

15.9 No reinterpretive rescue

The following are prohibited:

- replacing ARIS with any-positive-task status after external unblinding;
- dropping an opposed task-by-modality cell;
- rotating a modality to improve agreement;
- changing the recurrent session pair rule;
- changing $\delta_{\min}$, $Q_{\min}$, or the randomization threshold;
- treating technically invalid data as negative only when convenient;
- redefining the outcome as consciousness recovery;
- using a later landmark to claim that the day-28 hypothesis succeeded;
- interpreting a nonincremental association as a deeper hidden awareness result.

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16. Interpretation of Positive and Negative Patterns

16.1 ARIS positive with imagery-content support

This is the strongest pattern available within the protocol. It supports recurrent rule-sensitive responding plus neural activity compatible with both instructed imagery contents. It remains compatible with externally driven compliance, conditioned rule processing, and limited rather than autonomous agency.

16.2 ARIS positive without content support

This pattern supports a shared assignment-sensitive relation but not verified imagery. Generic preparation, inhibition, counting-like control, expectancy, or nonconscious rule processing remain viable.

The correct label is **content-unconfirmed recurrent rule-sensitive neural responding**.

16.3 Isolated task positive, ARIS negative

Possible explanations include:

- genuine task-specific command-related processing;
- preservation of one imagery system;
- modality-specific sensitivity;

- transient arousal;
- false positivity;
- failure of the external common representation.

The result may be clinically relevant but does not satisfy the cross-task claim.

16.4 ARIS positive, bedside behavior absent

This is compatible with cognitive motor dissociation, but it does not prove it. Alternatives include:

- residual rule-conditioned processing without conscious imagery;
- severe motor-expression failure;
- language or endpoint mismatch;
- later recovery beyond day 180;
- a prognostic false positive.

The participant’s care cannot be determined by ARIS alone.

16.5 Bedside command-following, ARIS negative

This is a false negative relative to the behavioral endpoint. It demonstrates that ARIS is not necessary for command-following. Likely causes include task demand, hearing, language, memory, lesion topology, poor neurovascular coupling, motion, or common-space failure.

A high false-negative rate in responsive controls invalidates the test before patient inference.

16.6 Resting network favorable, ARIS absent

This may reflect a permissive network without successful task implementation. It may also reflect a false-positive resting marker or an ARIS false negative.

16.7 ARIS favorable, resting network unfavorable

This would challenge a monotonic or global-network requirement. Sparse or reorganized routes might support rule-sensitive responding despite abnormal aggregate topology.

16.8 Positive active-control direction

A common direction spanning imagery and auditory counting supports generic instructed implementation. It may still predict command-following, but it weakens claims about imagery-specific control and agency.

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17. Philosophical Scope

17.1 Covert command-following as a defeasible inference

The phrase “covert command-following” is reserved for an inference supported by convergent evidence:

- intact enough auditory access;
- some receptive-language evidence;
- a mapping-sensitive assignment effect;

- recurrence;
- connected cross-task and cross-modal transport;
- absence of motion and passive-replay explanations;
- preferably, task-content support.

Even this conjunction does not provide direct access to the participant's private mental act. "Covert command-following capacity" is therefore a graded interpretation, not an operational synonym for ARIS.

17.2 Limited agency

Following an externally assigned rule can indicate conditional selection among alternatives. It does not establish autonomous goal formation, authorship, endorsement, or free choice. The participant does not choose the task, cue, timing, or response vocabulary.

The protocol therefore avoids describing every mapping-sensitive interaction as agency. Limited agency may be discussed only when language, content, recurrence, and negative-control evidence converge.

17.3 First-person structure

An instruction addressed to a participant has indexical form: it concerns what the addressee is to do. Successful implementation may be compatible with a representation linking the command to the participant's controlled activity. It does not entail a de se representation, phenomenal mineness, self-recognition, narrative identity, or self-consciousness.

Work on meditation and psychedelic states argues that self-consciousness is multidimensional and that superficially similar forms of self-loss can differ [11]. Those populations are remote from severe brain injury, so the citation supplies a conceptual caution rather than direct evidence about patients.

17.4 Temporal organization

The task requires retention of a mapping across a delay and recurrence across days. These are objective temporal properties. Theoretical work has emphasized nonzero conscious duration and proposed dynamic accounts of consciousness [7]. That theoretical source does not show that ARIS measures subjective duration or validate a specific neural clock.

17.5 Access and phenomenal consciousness

Higher-order theories propose that general cortical cognitive systems contribute to conscious experience [9]. Other theoretical work proposes evaluative or labeling functions for phenomenal properties [6]. ARIS does not directly test either proposal.

The protocol supports an asymmetrical inference:

- a robust positive result is evidence for a sophisticated public-test-compatible control relation;
- a negative result is weak evidence against consciousness because the task has many prerequisites;
- neither result settles phenomenal consciousness.

17.6 Report is not consciousness, and command-following is not report

Reliable bedside command-following is an overt access capacity. It is not necessarily communicative report, because a repeated movement may not convey unrestricted content. Communicative report is not conceptually identical to phenomenal consciousness. Conversely, absence of report does not prove absence of experience.

Consciousness research necessarily integrates conceptual, psychological, neural, and clinical questions [12]. The present protocol contributes one constrained empirical bridge; it does not reduce those domains to a classifier.

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18. Limitations

18.1 The external anchor may be identifiable but inadequate

The revised common space has fixed orientation and removes arbitrary relative rotations. It may still be a poor representation of cross-modal neural function. EEG-to-fMRI prediction can privilege event-locked or vascularly coupled signals, and severe injury may violate calibration relationships learned in controls.

This is the central unresolved technical risk.

18.2 Strict connected recurrence may be insensitive

Requiring all task-by-modality cells in two sessions reduces vulnerability to isolated effects but creates many false-negative pathways. One lesion-specific task failure, one unusable MRI session, or one weak coordinate family can determine the minimum.

The protocol accepts this trade-off for a high-specificity confirmatory claim. If sensitivity is inadequate in responsive injured controls, the study does not proceed.

18.3 The assignment effect is not uniquely intentional

The randomized interaction identifies an effect of cue meaning under a mapping context. It can still reflect conditional association, mapping retrieval, expectancy, preparation, effort, inhibition, or nonconscious rule-conditioned processing.

Active controls and content templates constrain these alternatives but cannot eliminate them.

18.4 Withholding cannot be independently verified

In noncommunicative patients, assigned withholding has no direct behavioral confirmation. Spontaneous imagery may occur during withhold trials, and some participants may not retain the instruction. Randomization reduces systematic bias but does not verify private compliance.

18.5 Active controls cannot guarantee equal effort

Silent counting matches several formal task demands but may be easier or harder than imagery for a particular injured participant. Pupillometry and autonomic measures are incomplete proxies for effort. A generic-demand explanation may remain even after adjustment.

18.6 Auditory and language checks are incomplete

Auditory evoked responses do not prove speech comprehension. Sentence-control responses do not prove understanding of the specific mapping. Aphasia, memory impairment, and fluctuating comprehension remain major false-negative sources.

18.7 Neurovascular coupling may distort fMRI

Regional hemodynamic changes can alter sign, latency, or amplitude. Even a finite-impulse-response model and vascular-reactivity assessment cannot fully recover neural activity from injured vasculature.

A failed cross-modal test may therefore reflect measurement physics.

18.8 EEG source localization remains uncertain

Individual head models and lesion masks reduce but do not remove uncertainty caused by abnormal conductivity, skull defects, cranioplasty, edema, and source leakage. The common anatomical coordinate system should not be mistaken for direct measurement of parcel-specific neuronal sources.

18.9 Diffusion and effective connectivity are proxies

Tractography is a model-dependent estimate, not direct evidence that a route communicates. Effective connectivity is conditional on a generative model, not observed causal transmission. Mechanistic terminology remains correspondingly qualified.

18.10 The primary cohort is selected

The MRI-complete day-28 risk set excludes participants who die early, remain unstable, cannot undergo MRI, or lack sufficient repeated data. These exclusions may enrich the cohort for medical stability and better prognosis.

The primary performance estimate cannot be generalized automatically to all patients with disorders of consciousness.

18.11 The endpoint is motor and language dependent

Reliable CRS-R command-following depends on auditory processing, receptive language, arousal, motor planning, and output. ARIS might predict restoration of any combination of those capacities rather than a new onset of phenomenal consciousness.

18.12 Day 180 is not a biological boundary

Some patients recover later. The fixed horizon is needed for prospective validation but does not define the end of recoverability. Longer follow-up is exploratory.

18.13 Prognosis occurs under observed care

Treatment, rehabilitation, complications, and care-limitation decisions affect the endpoint. The study does not identify untreated prognosis. Disclosure of neural findings can itself alter care and therefore outcome.

18.14 Site transport remains uncertain

A four-site external validation estimates performance in those sites. It does not establish universal transportability. Differences in referral, hardware, expertise, language, and treatment may remain.

18.15 A common direction is not a mechanism

Even if ARIS predicts outcome, a shared direction summarizes covariation. It does not explain how rules are represented, how imagery is generated, or why any neural state is experienced.

18.16 Simulation cannot replace clinical validation

Passing a planted-signal simulation shows that the algorithm can recover signals under specified assumptions. It does not show that patients instantiate those assumptions. Healthy and responsive controls are necessary but also do not guarantee validity in UWS or MCS−.

18.17 Prior-art and novelty limits

The likely contribution is the conjunction of cue reversal, an externally fixed cross-modal reference, a complete task-by-modality graph, cross-session recurrence, exact allocation-based inference, and locked prognosis. Multiple tasks and modalities are already motivated by prior patient work [4], and cross-session and cross-dataset distribution shifts are familiar in motor-imagery decoding research [31].

This manuscript does not claim discovery of the first consciousness axis. A systematic prior-art search comparing representational-similarity analysis, cross-decoding, hyperalignment, Procrustes methods, multiset latent models, longitudinal cognitive motor dissociation testing, and multimodal classifiers is required before any novelty statement is registered.

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19. Ethical Framework

19.1 Permission and participant welfare

Most target participants cannot provide conventional informed consent. Enrollment requires permission from a legally authorized representative under local regulation. Representatives will be told that:

- the study is investigational;
- positive findings do not prove phenomenal consciousness;
- negative findings do not prove its absence;
- the procedures may be tiring or burdensome;
- direct benefit is uncertain;
- withdrawal does not alter clinical care.

Behavioral or physiological distress triggers pause or termination.

19.2 No research-only sedation

Sedation will not be initiated solely to obtain research MRI or EEG. If clinically necessary sedation prevents interpretable testing, the session is postponed or recorded as missing.

19.3 Pain and comfort

Pain must be assessed and treated in disorders of consciousness [15]. ARIS does not measure pain. A negative neural result cannot justify withholding analgesia or comfort measures.

19.4 Clinical disclosure

Raw ARIS scores remain blinded during validation. Independently interpretable repeated active-paradigm findings may be reviewed by a multidisciplinary disclosure committee.

Disclosure requires:

- replication;
- artifact review;
- sensory and language review;
- explicit uncertainty;
- documentation that the result concerns neural instruction sensitivity rather than verified experience.

Clinical guidance supports multimodal integration rather than reliance on one neural test [3].

19.5 Life-sustaining treatment

ARIS must not serve as the sole basis for continuation, escalation, or withdrawal of life-sustaining treatment. The reasons include imperfect sensitivity, selected eligibility, uncertain phenomenological meaning, treatment feedback, and possible late recovery.

An unfavorable score is not evidence of futility.

19.6 Equity

The task may disadvantage participants with hearing impairment, language mismatch, pre-injury disability, atypical anatomy, limited MRI access, or lesions affecting one imagery system. Language history, translation, sensory checks, and modality eligibility will be reported by subgroup.

A technologically demanding biomarker can worsen disparities even if statistically accurate in selected centers.

19.7 Privacy and communication claims

Neural data may be identifying and may be misrepresented as revealing private thought. The protocol decodes only a narrow assigned task relation. It does not infer free thoughts, beliefs, preferences, pain, or consent.

Noninvasive brain–computer interfaces can support increasingly complex device control [44]. ARIS is not a communication channel. Communication would require deliberate trial-level selection, known error rates, stable confirmation, and message-level validation.

20. Reproducibility and Governance

The project will release, before external unblinding:

- task scripts;
- admissible randomization schedules;
- randomization seeds or auditable seed-generation procedures;

- common-space coordinate definitions;
- frozen modality encoders;
- containerized preprocessing;
- graph-generation code;
- unit tests for checkerboard rejection;
- simulation code;
- synthetic data;
- missingness and eligibility rules;
- prediction-model coefficients;
- an external-validation analysis container.

A leakage audit will test whether:

- patient contrast labels affected the modality orientation;
- held-out mini-blocks entered nuisance fitting;
- external outcomes entered feature or penalty selection;
- randomization occurred after rather than before label-adaptive processing;
- a post-command session entered a prebehavioral score;
- the favorable session pair was selected outside the null procedure;
- future information entered imputation;
- site recalibration used outcome information.

Any confirmed leakage invalidates the affected analysis.

Patient-level neural data will remain under controlled access because de-identification is imperfect. Derived features and executable code will be shared where consent and regulation permit.

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21. Response to Independent Review and Remaining Disagreements

21.1 Common-space identifiability

We agree with both reviewers that the former modality-specific orthogonal maps did not identify cross-modal cosine alignment. The earlier invariance proof applied only to a common transformation and was inapplicable to distinct EEG and fMRI maps.

The revised estimator uses one externally oriented fMRI-anchored reference and a uniquely specified EEG prediction objective. Patient-specific relative rotations, reflections, permutations, and sign choices are prohibited.

What remains unresolved is empirical adequacy. A fixed but poor cross-modal representation is still poor. We therefore do not claim that the objection is fully resolved until Stage 0 and control-cohort transport are successful.

21.2 Disconnected eligible pairs

We agree that the prior requirement that every pair differ simultaneously in task and modality created two disconnected parity classes and allowed a maximal checkerboard score.

The revised graph contains all six within-session edges among the four task-by-modality cells, plus leave-one-cell, leave-one-task, leave-one-modality, and cross-session tests. The primary statistic is governed by the weakest required component rather than a favorable average.

Finite-sample checkerboard rejection remains a mandatory simulation gate.

21.3 Null calibration

We agree that arbitrary trial-label permutation was invalid under mini-block assignment and temporal dependence. The primary probability now reconstructs the actual restricted mini-block randomization mechanism and reruns the complete adaptive pipeline.

We also agree that temporal shifts and phase surrogates are not automatically exact tests. They are now explicitly negative-control distributions. The normal-theory z threshold is removed, B is fixed, ties are conservative, and no null standard deviation is required.

21.4 Intentionality, imagery, and active control

We agree that a cue-by-mapping interaction is an assignment effect rather than proof of deliberate imagery, deliberate withholding, agency, or awareness. The terminology has been narrowed, an active nonimagery control has been added, and independent imagery-content templates are required for the stronger “imagery-supported” label.

A residual disagreement is preserved. We do not think the assignment effect is devoid of cognitive relevance merely because it is compatible with nonconscious rule processing. Recurrent reversal-sensitive responding across tasks, modalities, and sessions would still be a substantive neural capacity. We agree, however, that its intentional or conscious interpretation is defeasible.

21.5 Endpoint terminology

We agree that future CRS-R command-following validates prediction of bedside command-following, not recovery of consciousness. The title, abstract, primary estimand, and decision rules now use that endpoint consistently.

The manuscript retains a philosophical discussion of covert command-following, agency, and phenomenal consciousness because those inferences motivate the clinical question. They are no longer operational synonyms and are explicitly conditional on convergent evidence.

21.6 Temporal precedence

We agree that shuffling visit order was invalid. The revised analysis uses fixed injury-anchored landmarks, prior-only predictors, interval-censored lead time, competing events, and a secondary joint longitudinal-event model.

21.7 Prognostic model and unseen sites

We agree that setting an unknown logistic random intercept to zero is not a marginal prediction. The revised primary external prediction integrates over the estimated site-effect distribution. Because six development sites may estimate that distribution poorly, simulation and fixed-site sensitivity analyses are mandatory.

The high-dimensional baseline is now penalized, effective degrees of freedom are reported, and all model selection remains inside development resampling.

21.8 Mechanistic restraint

We agree that diffusion, source-resolved EEG, effective connectivity, and TMS were previously described too confidently. The revised manuscript treats diffusion as a proxy, effective connectivity as model-dependent, EEG localization as uncertain in lesioned heads, and fMRI as vulnerable to altered neurovascular coupling. Optional TMS is removed from the mandatory longitudinal composite and receives sham and sensory controls.

21.9 Confirmation and falsification

We agree that point-estimate success conflicted with upper-bound rejection. Confirmation now requires lower one-sided confidence bounds above both targets. Failure to confirm, strong evidence below target, and technical invalidity are separate result states.

21.10 Simulation evidence not yet available

The reviewers requested completed simulation and control evidence before preregistration. We agree. This manuscript specifies the required package but cannot report nonexistent results. It is consequently labeled pre-preregistration, and recruitment is prohibited until those results are available and the gates pass.

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22. Expected Contributions Under Different Outcomes

22.1 Confirmatory positive result

A confirmatory result would require:

- successful Stage 0 calibration;
- acceptable healthy and responsive-control sensitivity;
- low passive-replay positivity;
- checkerboard rejection;
- valid mini-block randomization inference;
- connected two-session recurrence;
- external lower-bound AUC above 0.80;
- external lower-bound incremental AUC above 0.10;
- acceptable site guardrails;
- no motion or cue-identity failure.

The justified conclusion would be that ARIS is a prognostic marker of later reliable bedside command-following in the defined population.

22.2 Prognostic but content-unconfirmed result

ARIS may predict command-following while imagery-content templates remain absent. This would support recurrent rule-sensitive responding as a prognostic marker but not a claim that both imagery tasks were consciously performed.

22.3 Prognostic but mechanistically unsupported result

ARIS may meet external criteria while the network-ordering hypothesis fails. The biomarker claim could survive, but the proposed frontoparietal–thalamic account would be rejected or left unsupported.

22.4 Mechanistically associated but nonincremental result

ARIS may correlate with alpha and thalamocortical measures yet add little beyond them. This would make it a redundant expression of broader recovery rather than the proposed incremental marker. The primary hypothesis would fail.

22.5 High specificity and low sensitivity

A rare ARIS pattern confined mostly to later command-followers could motivate a separate rule-in study. It would not rescue the current broad prognostic claim if the co-primary external criteria fail.

22.6 Null result

A technically valid null result could indicate:

- no stable common direction;
- excessive task demand;
- poor cross-modal transport;
- strong biological fluctuation;
- sufficient prognosis already contained in clinical and resting data;
- heterogeneity incompatible with one marker.

The appropriate response would be revision or abandonment of ARIS, not a claim that its failure reveals an inaccessible form of awareness.

22.7 Technical failure

If the common encoder, control sensitivity, randomization calibration, or checkerboard rejection fails, no biological conclusion follows. The protocol itself has failed.

23. Conclusion

Behavioral nonresponse after severe brain injury does not identify one cognitive or neural condition, but neither does an isolated neural activation establish covert awareness. A defensible test must separate physical cue identity from assigned meaning, distinguish generic preparation from task content, connect every task-by-modality cell, recur across sessions, respect the actual randomization unit, and predict a clearly named clinical endpoint in untouched sites.

The revised ARIS protocol replaces an unidentified cross-modal rotation with an externally fixed coordinate system and replaces a disconnected favorable average with a complete graph governed by its weakest required component. It uses exact allocation-based inference, treats temporal and spectral surrogates as diagnostics, requires two-session transport, and evaluates future reliable CRS-R command-following from fixed injury-anchored risk sets. Its prognostic model uses frozen penalization and true site-marginal prediction. Its mechanistic claims are separated from prognosis and stated in terms of model-dependent proxies and quantitative futility rules.

These revisions make the proposal formally more coherent, but they do not establish that it will work. The external common representation may be biologically inadequate, the strict graph may be insensitive, and assignment-sensitive activity may reflect generic or nonconscious rule processing. Simulation and control evidence must therefore precede preregistration and recruitment.

If those gates are passed and the external confidence-bound criteria are met, ARIS would support prediction of later bedside command-following from recurrent rule-sensitive neural responding. It would not prove phenomenal consciousness. If the gates or external criteria fail, the corresponding claim fails. That combination of operational precision, interpretive restraint, and permission for the method itself to be invalid is essential for ethically responsible research in behaviorally nonresponsive patients.

Source Records

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Peer Review Notes

- owen-bell: This is an unusually careful, ethically restrained, and genuinely falsifiable protocol. The manuscript correctly labels all numerical outcomes as predictions, and every in-text citation is substantively compatible with the supplied source record, although several cited records are preprints, theoretical papers, or protocols rather than confirmatory evidence. The principal defect is mathematical: the proposed TGEC statistic is not yet an identifiable measure of one common axis. Different modality-specific orthogonal maps can arbitrarily change cross-modal cosine alignment, while the strict cross-task, cross-modal pairing graph can yield a perfect score from two mutually opposed axes. The null calibration, temporal-precedence test, mechanistic model, and philosophical terminology also require revision before preregistration.
- elian-voss: This is a conceptually original, ethically careful, and unusually explicit protocol whose cited claims largely fit the supplied source records. It is not yet ready for preregistration because the cross-modal TGEC geometry is not

mathematically identified, several null and temporal tests do not have the claimed inferential validity, the validation endpoint measures recovery of overt motor command-following rather than awareness, and the external-validation and power plans remain under-specified. Major analytic revision and pilot or simulation evidence are required before the proposed score can support the stated conclusions.